Composite versus Direct: the problems, and what the numbers say

1 The setup

Every problem has the form

$$f(x) \;=\; g\big(h(x)\big), \qquad x \in [0,1]^6,$$

with h the *intermediate* (expensive, unknown) and g the *outer map* (cheap, known in closed form). Two surrogates compete:

	Observes	Models	Predicts f by
Direct	$f(x_i)$ only	a GP on f	reading off that GP
Composite	$h(x_i)$	a GP on each h_j	sampling h and applying g

Both get the same inputs, the same budget, the same initial design and the same acquisition function. Only the surrogate differs.

1.1 The intermediates

h is not an analytic test function. It is a draw from the *same* Gaussian process family the surrogates assume, built from random Fourier features:

$$h(x) = \sqrt{\tfrac{2}{M}} \sum_{m=1}^{M} a_m \cos(\omega_m^\top x + b_m), \qquad a_m \sim \mathcal{N}(0,1),\; b_m \sim U(0,2\pi),\; \omega_m \sim \mathcal{N}(0, \ell^{-2} I),$$

with $M = 1024$ and $\ell = 0.5$, then standardised so it has mean 0 and standard deviation 1.

This matters for the comparison. If h were something like Rastrigin, Composite might win or lose because a GP happens to model *that function* well or badly, and we would learn nothing about g. Drawing h from the surrogate's own prior removes that: Composite's model of h is correct by construction, so every remaining difference is attributable to the outer map.

Standardising also puts the mean at zero, so h takes both signs. Without that, h^2 would be injective over the region actually visited and the whole information question would be empty.

Range. Over $[0,1]^6$, h reaches about $[-4.3, +3.8]$: sd exactly 1, with 5% of the domain beyond $|h| > 2$ and 0.3% beyond $|h| > 3$. The range saturates — sampling a million points finds no more than sampling ten thousand — because it is a property of the function, not of the sampling. That is why root placements at ± 2 sit inside the bulk while a vertex at $h = 3$ sits at the edge of what is reachable.

Several intermediates. When $k > 1$, each h_j is an independent draw with its own seed ($1000 \times \text{draw} + j$); measured correlation between them is -0.10, i.e. zero. Maps sharing a draw seed share the same h_1, so switching g never changes the underlying functions.

Table 1: Every problem. k is the number of intermediates, $\bar{n}$ the measured average number of h values consistent with one observation, and the last column what g throws away. A dash in $\bar{n}$ means the fiber is continuous, so the count is infinite.

Problem	k	$g(h)$	$\bar{n}$	Bijective?	What g discards
`identity`	1	h	1.00	**yes**	nothing
`exp_b1`	1	e^h	1.00	**yes**	nothing (monotone, positive)
`sinh_b1`	1	$\sinh h$	1.00	**yes**	nothing (monotone, unbounded)
`shiftsq_c3`	1	$(h-3)^2$	1.01	**yes**	sign of h - c
`shiftsq_c0`	1	h^2	2.00	no	sign of h - c
`shiftsq_c0.5`	1	$(h-0.5)^2$	2.00	no	sign of h - c
`shiftsq_c1`	1	$(h-1)^2$	1.95	no	sign of h - c
`shiftsq_c2`	1	$(h-2)^2$	1.37	no	sign of h - c
`abs`	1	$\lvert h\rvert$	2.00	no	sign of h (with a kink)
`cos_w1`	1	$\cos h$	2.03	no	which of many preimages
`cos_w2`	1	$\cos 2h$	5.04	no	which of many preimages
`relu`	1	$-\max(h, 0)$	∞	no	everything below zero (censoring, away from the optimum)
`clip_c2`	2	$\mathrm{clip}(2h_1, -1, 1) + h_2$	—	no	the clipped coordinate inside the saturated regions
`sum`	2	$h_1 + h_2$	—	no	how the sum splits (Truss / RCM40)
`product`	2	$h_1 h_2$	—	no	how the product factors, and sign (DTLZ2)
`norm`	2	$\sqrt{h_1^2 + h_2^2}$	—	no	direction (Color)
`max`	2	$\max(h_1, h_2)$	—	no	the smaller value (Lake)
`ratio`	2	$\frac{\sigma(h_1)}{0.05+\sigma(h_1)+\sigma(h_2)}$	—	no	scale (RGB)

2 What $\bar{n}$ means

$\bar{n}$ is the average number of h values that could have produced the y you observed:

$$\bar{n} = \mathbb{E}\big[\,|g^{-1}(y)|\,\big].$$

$\bar{n} = 1$ means y determines h — the map is injective, and observing h tells you nothing new. $\bar{n} = 2$ means a typical observation leaves two candidates.

It is *measured*, by counting crossings of $g(z) - y$ over the range h actually reaches, not inferred from the algebra. The two come apart: `shiftsq_c3` is a parabola, and so two-to-one on paper, but its vertex sits at $h = 3$ where h almost never goes, so in practice $\bar{n} = 1.01$ and it behaves as a bijection.

The same measurement needs care with flat regions. $-\max(h, 0)$ sends the entire half-line $h \le 0$ — about half the domain — to a single value, so those observations have an *infinite* preimage set. A crossing counter reports 1 there, because a flat region never crosses. `relu` is therefore marked ∞ and grouped with the non-bijective maps.

3 Results

Table 2: Bijective maps: final $\log_{10}$ regret over 20 runs, mean ± SD. Gain is Direct − Composite, so positive favours Composite. p is a one-sided paired t-test.

Problem	Direct	Composite	Gain	p
`identity`	-1.789 ± 1.323	-2.198 ± 1.176	$+0.409 \pm 1.138$	0.062
`exp_b1`	-2.868 ± 0.378	-3.493 ± 1.474	$+0.625 \pm 1.309$	0.023
`sinh_b1`	-0.490 ± 1.133	-0.933 ± 1.105	$+0.444 \pm 1.126$	0.047
`shiftsq_c3`	-4.824 ± 1.282	-6.571 ± 1.762	$+1.747 \pm 1.482$	<0.001

Table 3: Non-bijective maps, same convention.

Problem	Direct	Composite	Gain	p
`shiftsq_c0`	-5.650 ± 1.306	-7.665 ± 0.580	$+2.015 \pm 1.454$	<0.001
`shiftsq_c0.5`	-5.510 ± 1.134	-7.824 ± 0.284	$+2.313 \pm 1.041$	<0.001
`shiftsq_c1`	-5.594 ± 1.191	-7.797 ± 0.445	$+2.203 \pm 1.181$	<0.001
`shiftsq_c2`	-5.494 ± 0.899	-7.927 ± 0.133	$+2.433 \pm 0.893$	<0.001
`abs`	-2.746 ± 0.575	-3.093 ± 0.582	$+0.347 \pm 0.762$	0.028
`cos_w1`	-4.429 ± 2.566	-6.497 ± 1.863	$+2.068 \pm 2.497$	<0.001
`cos_w2`	-4.647 ± 0.907	-6.820 ± 0.804	$+2.173 \pm 1.166$	<0.001
`relu`	-1.303 ± 1.150	-1.847 ± 1.283	$+0.545 \pm 1.600$	0.072
`clip_c2`	-1.166 ± 0.491	-2.964 ± 2.285	$+1.798 \pm 2.103$	<0.001
`sum`	-1.884 ± 1.208	-2.230 ± 1.128	$+0.346 \pm 1.149$	0.097
`product`	-1.167 ± 1.125	-1.378 ± 1.082	$+0.211 \pm 1.109$	0.203
`norm`	-1.885 ± 0.355	-2.174 ± 0.218	$+0.290 \pm 0.276$	<0.001
`max`	-1.092 ± 0.507	-1.706 ± 0.626	$+0.615 \pm 0.725$	<0.001
`ratio`	-2.172 ± 1.120	-2.280 ± 1.364	$+0.109 \pm 0.820$	0.280

3.1 How to read these

The metric. Simple regret, $R_T = f_{\text{best}}(T) - f^\star$, after $T = 45$ evaluations, reported as $\log_{10} R_T$. Lower is better. Because it is a log, the *gain* is a ratio: a gain of $+2$ means Composite's final regret is 100 times smaller.

The spread. Quoted as $\pm$ one standard deviation across the 20 runs, not a standard error — it describes how much a single run varies, not how precisely the mean is known. The standard error is roughly $1/\sqrt{20}$ of it.

The p-value. A one-sided paired t-test, asking whether Composite's mean regret is lower. Paired because both arms share an initial design within each run, so the comparison is run by run and the large variation between designs cancels.

A caution. These distributions are skewed — several standard deviations exceed their own means — so a t-test can be carried by a few large wins. On `relu`, Composite actually loses 11 of 20 runs while still posting a positive mean gain. A p-value near 0.05 on a row with a large SD should be read as suggestive rather than settled.

The -8 floor. $\log_{10}$ regret is clamped at -8. Four `shiftsq` rows have Composite at -7.7 to -7.9, so they are pressed against it and their gains are lower bounds. The reference optimum is known to about 10^{-15}, so the clamp is discarding real resolution and should be lowered.

4 Convergence

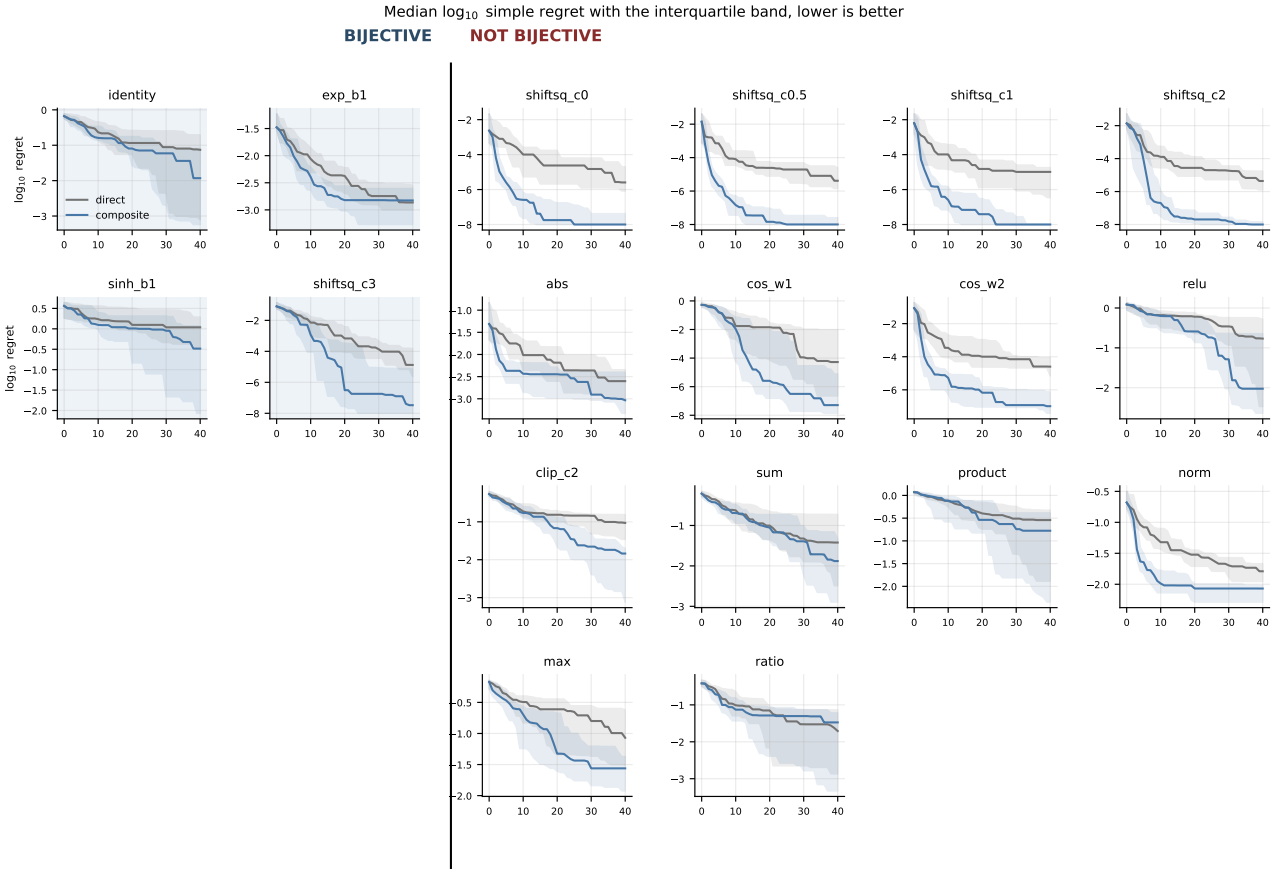


Figure 1: Median $\log_{10}$ regret against evaluation, interquartile band shaded. Bijective maps left of the rule, non-bijective right. Composite wins throughout *both* sides, which is the central observation: the advantage is not information, because on the left there is none to be had.

5 Why Direct loses even when nothing is hidden: z^3

`mono3`, $g(h) = h^3$, is strictly increasing. It is bijective, so observing h conveys nothing that y does not already determine. Composite should have no advantage — and it has a large one. A plain regression experiment, with no optimisation anywhere, shows why.

Eighty training points, twenty held out, 30 fits per map. RMSE is divided by the standard deviation of whatever is being predicted, so 1.0 means no better than guessing the mean.

Table 4: Held-out RMSE, normalised. "Fitted" means both models estimate their hyperparameters by maximum likelihood, so neither is given anything the other lacks.

	Direct	Composite (fitted)	
Problem	on f	on h (inner)	on f (outer)
`identity`	0.588	0.588	0.588
`mono3` $= h^3$	**1.071**	0.588	**0.817**
`root3`	1.287	0.588	0.962
`shiftsq_c0` $= h^2$	1.056	0.588	0.865

Three things to read off it.

The inner RMSE is the same in every row, 0.588**.** It must be: h is literally the same function in all four, and Composite always fits the same GP to it. A useful check that nothing else is varying.

Direct scores above 1.0 **on all three nonlinear maps.** It is worse than predicting the mean of f. With eighty training points, a stationary GP placed on h^3 has negative skill.

Composite, on the same eighty points with the same fitted hyperparameters, scores 0.817**.** Nothing is being hidden from Direct: h^3 is bijective, so y determines h exactly. Both models estimate their own kernel. The only difference is that one models h and applies the known cube afterwards.

So Direct's failure is not a shortage of data and not a worse fit. It is that $f = g(h)$ is *not a Gaussian process* when g is nonlinear, and a GP placed directly on f is the wrong shape of model. That is misspecification, isolated from information, in one table.

`identity` is the control: 0.588 against 0.588, identical to three figures, because there $f = h$ and the two models are the same model.

6 What this means overall

- **Composite wins on nearly every problem**, bijective or not, usually by one to two orders of magnitude in final regret.
- **Bijectivity does not predict the gain.** `shiftsq_c3` is effectively bijective and gains +1.75; `abs` is two-to-one and gains +0.35. The two groups overlap heavily.
- **The reason is that most of the advantage is not information.** When g is nonlinear, f is not a GP and Direct is fitting the wrong model. Bijectivity removes the information channel and leaves that one untouched.
- **The quantity that does track the gain is how low the problem can be driven at all** ($r = -0.93$ against Composite's final regret). Where g is flat at its optimum there is a broad near-optimal plateau and the gap can be enormous; where g is steep there, neither arm gets far and the gap is small.

Left out. `cos_w4` is excluded pending a closer look. It has the most preimages of any map ($\bar{n} = 9.75$) and nearly the smallest gain, because at $\omega = 4$ the posterior uncertainty in h spans most of a half-period and Composite's predictive on f becomes as wide as f itself — it knows h and that tells it nothing about the objective.

One caveat on these tables. Composite here is given the generator's true kernel, while Direct must estimate one. On `identity`, where no misspecification is possible, that alone is worth the entire +0.409 gain; making Composite fit its hyperparameters drops it to +0.206. The nonlinear rows keep their large gains either way, but the bijective rows should be read as upper bounds until the all-fitted grid is run.